

Fresh Adversarial Review and One-Round Triage

Benchmark Stream Construction, Not Attack Prevalence, Produces the Regime Structure of Streaming Intrusion Detection

Review date: 26 August 2026

Recommendation: MAJOR REVISION - DO NOT SUBMIT THIS VERSION

Core value: exact same-full-multiset construction audit.

Blocking defects: central estimand, capped-regime method identity, and companion-result disclosure.

Target: *Benchmark Stream Construction, Not Attack Prevalence, Produces the Regime Structure of Streaming Intrusion Detection*

Audience: Author, before submission to ACM Digital Threats: Research and Practice (DTRAP)

Review date: 26 August 2026

Scope and direct answer

This is a fresh hostile-referee review of the 10-page manuscript supplied as manuscript_review(1).pdf. It applies the five requested DTRAP criteria and then executes the supplied one-round triage rule once. The review checked the manuscript against the supplied reviewer prompt, triage template, and response shelf; the official DTRAP scope and author guidance; the cited Catillo, Arp, Cao, and Adams-MacKay papers; current conformal-prediction literature; and the public CALIBURN preprint.

The code, manifests, claim ledger, corrected-incident log, and paper source were not included in the review packet. Claims that depend on those materials are therefore not independently verified here. A manifest can prove lineage and integrity; it cannot by itself prove that code, labels, preprocessing, or scientific interpretation are correct.

Direct answer: do not submit this version. Major revision is required. The exact-permutation contrast is useful, the dilution arithmetic is clear, and the manuscript is unusually candid about limitations. But the paper currently asks that evidence to carry claims it cannot carry. The rank inversion is measured on test sets with only 32.5% record overlap and under a label-privileged ECOD protocol; the design changes test prevalence and sample membership together, so the title's "not attack prevalence" contrast is not identified; the method-identity argument is proved below the run-length cap even though the evaluated system spends almost all of its run above that cap; and the public companion paper contradicts the manuscript's assertion that the two papers report disjoint result sets.

Relevance	ADEQUATE	Clearly in DTRAP's evaluation-practice scope, but practitioner consequences are asserted more than measured.
Novelty or utility	ADEQUATE	The exact same-multiset measurement and audit are useful; the conceptual warning about order, sampling, and temporal snooping is established prior art.
Internal validity	FAILING	The central causal framing, method-rank inference, method-identity inference, and metric interpretation have material unresolved defects.
External validity	WEAK	The manuscript states many limitations, but the title, abstract, and conclusion still generalize far beyond two selected benchmarks and one split.
Transparency	FAILING	The provenance design is plausible but unaudited in this packet, the methods are not self-contained, and the "disjoint result sets" disclosure is contradicted by the public companion preprint.

1. Relevance - ADEQUATE

- 1. The topic is within DTRAP's scope.** DTRAP says it targets prevention, identification, mitigation, and elimination of digital threats and aims to bridge academic research and industry practice. Its author guidance asks research papers to use scientific rigor and structured observation to illuminate an advance. A paper showing how benchmark construction changes the apparent behavior of intrusion detectors is therefore germane. The manuscript's concrete recommendation in Section 10 - report natural-order results beside constructed results, or identify the operating point as a property of assembly - is directly relevant to evaluation practice.
- 2. The operational-security contribution is thinner than the framing suggests.** The manuscript says it matters to people evaluating streaming IDS, but it reports no alert volume, false-positive burden at an operator-selected point, throughput, memory cost, or real deployment outcome. Section 3.4 explicitly says there is no wall-clock instrumentation, and Section 9 says, "Nothing here estimates deployment prevalence or deployment performance." That is honest, but it means the paper is an evaluation-methodology paper for researchers, with only indirect practitioner utility. DTRAP can publish that lane, but the introduction should stop implying that it directly evaluates operational detection.
- 3. The prevalence of the alleged practice is unsupported.** The abstract opens, "Streaming intrusion-detection results are almost always reported on benchmark streams that were assembled," and Section 4.3 says, "The moderate regimes the literature reports on these benchmarks are products of the constructions examined next." The paper contains no systematic literature sample, coding protocol, or table of prior studies showing how often day interleaving, capture pooling, or round-robin replay is used. The only construction demonstrably tied to a prior evaluation is the authors' own archived earlier evaluation. "Almost always" and "the literature reports" are factual frequency claims, not rhetoric. Add a review of the relevant studies or narrow the statements to "some evaluations, including the earlier evaluation audited here."
- 4. "Natural order" is not automatically the operational ground truth.** CICIDS2017 is a controlled five-day scenario, not a production network; the manuscript itself uses only 76.628% of eligible flows. LITNET's selected captures have no coherent global chronology. The useful practitioner lesson is therefore not "chronology is always correct." It is "preserve and disclose capture structure, and test the sensitivity of conclusions to justified constructions." That claim is both more defensible and more useful.

2. Novelty or utility - ADEQUATE

1. **The conceptual warning is known art.** Section 2 says the contribution is "orthogonal" to prior benchmark criticism. That is too strong. Catillo et al. explicitly warn that random partitioning processes timestamped intrusion records "in an arbitrary order," can alter split representativeness, and is especially problematic for drift analysis. Arp et al. separately identify sampling bias, mixing incompatible sources, and temporal snooping as common security-ML failures. The present paper does not discover that temporal assembly is non-neutral.
2. **The paper does exceed that prior art in one useful way.** Catillo and Arp provide the general diagnosis. Cao et al. show that streaming-anomaly evaluation is sensitive to anomaly types, drifts, and assessment design, including the mistake of ranking scores from changing models over a whole stream. None of those sources, in the material checked, provides this manuscript's exact CICIDS2017 intervention: the same 1,600,000-record multiset under timestamp order versus day-round-robin order, with overlap, relocation, prevalence, and deterministic rank effects reported together. That is the paper's real contribution. Call it a quantified case study and mechanism audit, not a general discovery that construction rather than prevalence creates streaming-IDS regimes.
3. **The LITNET pooling identity is arithmetic, not novelty.** Section 5.4 correctly says that 6.4982% is the equal-weight mean of 15.7747%, 3.544%, and 0.176% because equal 75,000-record test tails are pooled. That is a useful audit check, but it is a tautology produced by equal budgets and a divisible round-robin boundary. Its contribution is pedagogical utility: it exposes how a pooled operating point can have no corresponding capture. Do not present the identity itself as a scientific discovery.
4. **The method-identity finding is potentially valuable, but it is currently housekeeping with an overextended conclusion.** Discovering that released code does not implement the described change-point score is necessary correction work. It becomes a general contribution only if the paper states and proves a reusable condition, shows the result over the algorithm's actual operating regime, and demonstrates its effect on the final score. The current algebra concerns $P(r_t = 0)$ below the cap, while the deployed score uses $P(r_t \leq 5)$ and the evaluations spend nearly all their time at or beyond the cap. As written, the paper has found a serious implementation defect; it has not yet established a complete general characterization of the evaluated detector.
5. **The companion paper materially overlaps the reported results.** The manuscript says, "The two papers report disjoint result sets" (Companion Paper Disclosure, p. 9). The public CALIBURN preprint reports the same rounded LITNET pooled results (CALIBURN 0.943, HST 0.261, ECOD 0.229) and the same rounded CICIDS interleaved results (CALIBURN 0.545, HST 0.433, LODA 0.342, ECOD 0.419, LOF 0.863). Those correspond to Tables 3 and 4 here at higher precision. If CALIBURN is the suppressed companion - the exact datasets, construction, score lineage, and numbers make that identification compelling - the disjoint-result claim is false. The new paper may still be novel as an audit and construction contrast, but it must state exactly which measurements are reused, re-derived, corrected, or new.

3. Internal validity - FAILING

1. **"On identical records" is materially misleading for the ranking claim.** Section 5.1 verifies an exact permutation of the full 1,600,000-record multiset. Section 5.2 then states that the two held-out slices share only 78,000 of 240,000 records, or 32.5%. Therefore the deterministic rank inversion in Section 5.3 is not measured on an identical held-out sample. Training membership, validation membership, test membership, test prevalence, and within-split order all change. The defensible statement is: "The same full-stream multiset, repartitioned by reordering under one positional 70/15/15 split, produces different held-out samples and a different within-sample ordering of the two deterministic scores."
2. **The title's construction-versus-prevalence contrast is not identified.** Reordering is the upstream intervention, but it changes held-out prevalence from 68.235% to 25.2396%. It also replaces 67.5% of the held-out records. The experiment therefore shows that construction changes the evaluation regime through prevalence and sample composition; it does not show that construction, "not attack prevalence," produces the regime. Holding whole-stream prevalence constant is not the same as holding evaluation prevalence constant. To exclude prevalence as an explanation for the rank effect, the paper needs a factorial or matched design: comparable held-out records or prevalence across constructions, plus separate interventions on training order, test order, and split membership.
3. **The paper has not isolated the mechanism of the method-rank inversion.** Section 5.2 identifies dilution as the mechanism for the prevalence change, not for the algorithmic rank change. The rank inversion can arise from different test membership, different attack-family composition, different training contamination, online adaptation to ordering, or ECOD's training advantage. Calling the inversion the paper's "central evidence" (Section 5.3) is acceptable only for the narrow pipeline-level fact that the evaluation outcome changes. It is not evidence that ordering itself changes detector quality on the same test population.
4. **ECOD makes the central comparison scientifically unfair.** Section 5.3 and Section 9 admit that ECOD is fit on benign-only training data and is "label-privileged" relative to the unsupervised streaming methods. This is not a footnote-sized nuisance. The central rank inversion is ECOD versus the detector. Where ECOD wins, supervision is an uncontrolled explanation; where it loses, a static benign-only model is being compared with a model that updates through a mixed stream under a protocol the paper does not fully specify. Either align information access and update rules or present ECOD only as a diagnostic reference and remove the method-superiority language.
5. **Determinism does not remove sampling uncertainty.** The natural-arm margin is $0.755142 - 0.728337 = 0.026805$ AUC-PR. Calling one method a winner because both implementations are deterministic addresses seed variance only. It does not address uncertainty from the selected time window, correlated attack runs, subsampling, or finite events. No confidence interval, event-block bootstrap, rolling-origin estimate, or repeated split is reported. With a median/p90/max attack-run length of 2/70/2522, an IID row bootstrap would also be questionable. The claimed inversion needs uncertainty that respects time or attack events.
6. **The fixed split is load-bearing, not peripheral.** Section 5.1 says no split-rule sensitivity analysis was run. Section 9 says the natural test is one 204.2-minute Friday-evening window and that the effect operates through the attack schedule meeting a fractional tail split. A single split point can manufacture the measured contrast just as surely as the interleaving can. At minimum, evaluate multiple chronological cut points or rolling origins and report when the inversion survives. Until then, the result is "at this split," not a stable property of CICIDS2017.
7. **"True timestamp order" overstates what was evaluated.** Section 4.1 retains only 76.628% of eligible CICIDS records, with day-specific retention from 64.399% to 93.156%. Sorting a stride-subsampled corpus preserves relative order but not the full event stream. Missing flows can break attack runs, change online sufficient statistics, and alter adaptation. The paper measures only the prevalence bias (-2.1464 points); it does not measure changes in attack-family mix, feature distributions, run lengths, or detector results caused by the budget. Use "timestamp-sorted budgeted subsample" and either analyze these distortions or remove claims that rely on full natural chronology.
8. **The experimental protocol is not self-contained.** The paper does not specify the feature set, scaling, categorical handling, missing/infinite-value treatment, eligibility rule for the 2,087,997 records, per-day budgets, stride rule, warm-up, hazard and cap values in the main method description, online update timing, whether held-out labels ever

affect updates, baseline hyperparameters, validation selection, or exact metric implementation. The public companion paper contains some of these details, which proves they can be stated. A manifest is not a substitute for a readable method. A referee cannot determine whether the streaming and batch methods receive comparable information.

9. **"AUC-PR" is undefined at the point where tie handling matters most.** Section 5.3 says the chance floor equals prevalence, which is the standard interpretation for average precision or the expected precision of an uninformative ranking. But the paper never says whether it computes scikit-learn average precision, trapezoidal area under the precision-recall curve, or another convention. These are not interchangeable, especially in Table 5 where 92.74% of repaired scores tie at exactly 0.25. Name the function, version, positive class, interpolation/tie convention, and sample weighting.
10. **The tail-term interpretation does not follow from the reported diagnostic.** Section 3.1 says the change-point branch binds on 75.039% of post-warm-up records but contributes a mean score of 0.002531, then concludes, "The discriminative work is done by the tail term throughout." AUC metrics depend on ranking, not absolute score magnitude. A small-valued branch can still determine ranking; in fact, "binds" means it sets the final max score on most records. The paper needs tail-only, branch-only, and combined AUC results, or rank correlations and label-conditional distributions. The mean alone cannot establish which term discriminates.
11. **The method-identity proof does not cover the operating regime used for evaluation.** Section 3.2 proves $P(r_t = 0)$ equals the hazard below the run-length cap, then admits that for $t \geq 500$ truncation removes growth mass, the posterior wanders (mean 0.01099, max 1), and every evaluated stream spends nearly all of its length in that regime. The final score also uses $P(r_t \leq 5)$, not $P(r_t = 0)$. The paper therefore has not proved that the auxiliary is data-independent or non-detecting where almost all reported scores are generated. Characterize the truncated recurrence and the scored short-run mass, not only the untruncated reset probability.
12. **The repair experiment tests one untuned repair, not "the textbook repair."** Section 7 uses one vague Normal-Inverse-Gamma prior predictive, one `udp_flood` stream, a 200,000-record prefix, and a 30-minute cap. The paper correctly says it draws no conclusion about a tuned prior-predictive scale, but the abstract and Contribution 3 still say "the textbook repair" saturates and that both variants are degenerate. Replace this with "one untuned prior-predictive repair under the tested configuration." A meaningful corrected-BOCPD result requires a training/validation tuning protocol and more than one stream.
13. **The statement "AUC-ROC of 0.50962 means constant" is false.** Table 5 reports 747 distinct repaired scores and standard deviation 0.159341, so the score is not constant. An AUC-ROC near 0.5 indicates near-random class ranking on that sample; it does not identify a constant scoring function. Without uncertainty, even "chance" should be phrased as "close to the 0.5 chance benchmark." This is a mandatory wording correction.
14. **The exchangeability language is statistically wrong.** Section 2 says constructions that reorder records to balance splits "manufacture the exchangeability" conformal methods assume, and Section 4.3 titles the natural stream "not exchangeable." Reordering one realized finite dataset does not establish that observations are exchangeable; exchangeability is a distributional symmetry under permutation. Attack runs are evidence of serial structure, not by themselves a proof about the joint distribution. Moreover, conformal methods beyond exchangeability and non-exchangeable conformal risk control already exist. Say that interleaving can mask temporal dependence or make exchangeability look more plausible to a diagnostic; do not say it manufactures the assumption.
15. **The prevalence sweep does not isolate prevalence either.** Section 6 changes prevalence by stratified resampling, and explicitly says only the unresampled draws see the same records. Record identity, repetition, attack-family mix, and temporal adjacency can therefore change with the nominal prevalence level. The reported gradient is a property of those resampled datasets, not a clean prevalence intervention. In addition, additive lift $AP - prevalence$ has a prevalence-dependent ceiling of $1 - prevalence$, so its scale is not constant across levels. Report uncertainty and a normalized measure such as $(AP - p)/(1 - p)$, while retaining raw AP and p .
16. **Three seeds do not justify distribution language.** Section 5.5 correctly avoids flat HST rankings, but means and standard deviations over three seeds are fragile summaries. Table 4 reports only means over three resampling draws and omits ranges or standard deviations while claiming "small spread." The paper should print the three values or at least range and standard deviation wherever seed or resample sensitivity is discussed.

4. External validity - WEAK

- 1. The manuscript's limitations section is stronger than its title and conclusion.** Section 9 says, "Two benchmarks, one split rule," and "Nothing here estimates deployment prevalence or deployment performance." Section 10 nevertheless says, "On these benchmarks, the regime structure that motivates streaming IDS evaluations ... is manufactured by the step that assembles captures into streams." The paper evaluated one exact CICIDS construction and one algebraic LITNET pooling construction. It did not establish what generally motivates streaming-IDS evaluations or what the literature as a whole reports.
- 2. The CICIDS effect is schedule-specific.** The reported dilution occurs because only Friday contributes held-out attacks while Monday through Thursday contribute 162,000 attack-free rows under day round robin. That mechanism is real and useful, but it depends on CICIDS2017's scripted day schedule and the selected split boundary. It does not establish the direction or size of construction effects on another dataset, another split, or an operational stream.
- 3. The LITNET result is a selected compositional example.** Three attack types, equal 500,000-record budgets, and equal 75,000-record tails guarantee the prevalence mean. The paper does not show that other capture selections, weights, or split boundaries behave similarly. Its general lesson is that pooled prevalence depends on mixture weights; the numeric 6.4982% and the observed method values do not generalize.
- 4. Chronology is not a universal gold standard.** In multi-sensor or multi-site systems, an operational stream may legitimately merge asynchronous sources. In public testbeds, timestamp order can preserve scenario artifacts that are themselves nonrepresentative. The externally valid requirement is to justify the stream model, preserve source and time metadata, and report sensitivity to plausible constructions. The manuscript currently risks replacing one unexamined convention with another.
- 5. The paper buries its most generalizable contribution.** The durable finding is methodological: stream construction is a treatment that belongs in the experimental design and provenance ledger. Report mixture weights, capture identity, split membership, overlap, and natural/constructed sensitivity. That principle can generalize beyond these benchmarks without claiming that all moderate regimes or rankings are manufactured.

5. Transparency - FAILING

- 1. The described provenance system is credible as engineering, but not yet audited evidence.** Section 8 lists git commits, configuration hashes, input SHA-256 values, seeds, environment hashes, timestamps, output paths, build failures on orphan numbers, and a claim ledger. Those controls can prevent transcription drift and map prose to runs. They do not establish that the run used the right records, labels, code, metric, or interpretation. The artifact was not supplied in this packet, so the claims "every number" and "verified mechanically" remain unverified here.
- 2. The paper relies on the artifact to supply essential methods.** Reproducibility artifacts complement a paper; they do not relieve the paper of specifying what was done. The omissions listed in Internal Validity finding 8 prevent substantive review even if every displayed number has a manifest. Add a compact protocol table and exact stream-construction pseudocode to the paper.
- 3. The correction confession is not specific enough for the severity involved.** The introduction says earlier versions used a composite construction and described a scoring rule the code did not implement. Those are not minor errata; they affect the identity of the method and the interpretation of the headline results. Give version identifiers and dates, state whether the versions were public, submitted, or accepted, identify which claims and tables were invalidated, and link each to a corrected incident. "Eighteen numbered incidents" without severity or closure status is not an adequate correction history in the paper.
- 4. The companion disclosure contains a checkably false sentence.** "The two papers report disjoint result sets" conflicts with the public CALIBURN Tables 4 and 5 and Figure 5, which report the same rounded LITNET pooled and CICIDS interleaved values reproduced here at higher precision. Replace the sentence with an explicit overlap matrix: reused measurement, old interpretation, corrected interpretation, new measurement, and correction status.
- 5. The anonymous-review story needs editorial handling.** DTRAP requires double-anonymous submissions. The exact datasets, unique results, correction narrative, and public companion artifact make the author and companion readily identifiable. Suppressing the reference while making a false disjointness claim is worse than a properly blinded self-citation and confidential note to the editor. Follow DTRAP's instructions for prior appearance and give the editor the companion DOI, arXiv version, correction status, and overlap statement.
- 6. The AI disclosure is not the problem.** The manuscript discloses use of large language models for scripts, prose, and an adversarial audit and assigns human responsibility. Current ACM authorship policy no longer requires disclosure of writing assistance, but voluntary disclosure is acceptable. The review concern is verification: any LLM-written method or claim must be checked against code and artifacts, exactly as the manuscript says. Do not let the disclosure distract from the concrete overlap and method-description defects.

Overall recommendation

MAJOR REVISION. The paper has a publishable core, but this version should not be submitted. The three findings that drive the recommendation are:

1. The central construction/ranking claim is not identified as written: the held-out samples differ by 67.5%, test prevalence changes by 42.9954 points, ECOD has privileged label access, and no split or sampling uncertainty is reported.
2. The method-identity conclusion is not established over the regime that produced almost all reported scores: the proof is below the cap, the runs are mostly beyond it, and the final score uses $P(r \leq 5)$, while the branch that allegedly contributes no signal sets the score on 75.039% of records.
3. The transparency statement that the companion papers have disjoint result sets is contradicted by the public CALIBURN preprint.

Single strongest sentence: "The construction effect is measured under a fixed 70/15/15 chronological split; no split-rule sensitivity analysis exists, and the CICIDS effect operates through per-day attack scheduling meeting a fractional tail split." (Section 9, p. 8.) It is precise, falsifiable, and correctly limits the inference.

Single weakest sentence: "The two papers report disjoint result sets." (Companion Paper Disclosure, p. 9.) Publicly available tables show otherwise.

One-round triage

The table below applies the supplied VERIFY / FIX / SHELF rule once. "FIX" means a manuscript or disclosure change that follows directly from verified evidence. "SHELF" means a judgment item or new analysis whose cost cannot be established from the supplied packet. No manuscript source or project tree was supplied, so this review cannot execute code, measure revision-window runtime, regenerate macros, or run the provenance gate.

Topic is within DTRAP scope.			SHELF as prepared scope response; no paper change required.
Class Judgment grounded in policy	Verification VERIFIED-TRUE: DTRAP scope and author guidance expressly include rigorous research on digital-threat identification and practice.	Disposition SHELF as prepared scope response; no paper change required.	
"Almost always" and "the literature reports" are unsupported frequency claims.			FIX
Class Factual	Verification VERIFIED-TRUE: no literature sample, coding method, or prior-study table appears in the manuscript.	Disposition FIX: weaken or add systematic evidence.	
Prior art already warns about arbitrary temporal ordering and temporal snooping.			FIX
Class Factual	Verification VERIFIED-TRUE: Catillo pp. 3-4 and Arp Sections 2.1-2.2 state these risks.	Disposition FIX: revise novelty positioning.	
The CICIDS arms are an exact full-multiset permutation and the headline arithmetic is internally consistent.			KEEP.
Class Factual	Verification VERIFIED-TRUE: $352,962/1,600,000 = 22.060125\%$; $163,764/240,000 = 68.235\%$; $60,575/240,000 = 25.2396\%$; difference = 42.9954 points; overlap = $78,000/240,000 = 32.5\%$.	Disposition KEEP.	
"On identical records" is misleading for method comparison.			FIX
Class Factual	Verification VERIFIED-TRUE: Section 5.2 says only 32.5% of held-out records overlap.	Disposition FIX: say identical full-stream multiset, different held-out samples.	
"Not attack prevalence" is not identified.			FIX framing now; SHELF factorial experiment.
Class Factual design claim	Verification VERIFIED-TRUE: held-out prevalence is changed by the intervention; no prevalence-matched cross-construction contrast exists.	Disposition FIX framing now; SHELF factorial experiment.	

	ECOD's information advantage confounds the central inversion.		FIX comparison language now; SHELF protocol-aligned rerun.
Class Factual	Verification VERIFIED-TRUE: Sections 5.3 and 9 call ECOD benign-only and label-privileged.	Disposition FIX comparison language now; SHELF protocol-aligned rerun.	
	Determinism does not establish a stable winner.		SHELF until compute cost is measured; do not state a stable winner meanwhile.
Class Factual/statistical	Verification VERIFIED-TRUE: no interval, repeated split, block bootstrap, or event-level uncertainty is reported.	Disposition SHELF until compute cost is measured; do not state a stable winner meanwhile.	
	Split sensitivity is absent and the natural test is one 204.2-minute window.		SHELF; existing S5 response acknowledges it but does not cure the inference.
Class Factual	Verification VERIFIED-TRUE: Sections 5.1 and 9 state both facts.	Disposition SHELF; existing S5 response acknowledges it but does not cure the inference.	
	The paper is not methodologically self-contained.		FIX
Class Factual	Verification VERIFIED-TRUE by manuscript inspection; core preprocessing, eligibility, update, hyperparameter, and metric details are absent.	Disposition FIX: add compact protocol table and pseudocode.	
	Mean branch magnitude does not prove the tail term does the discriminative work.		FIX claim; SHELF ablation if runtime is over 30 minutes.
Class Factual inference	Verification VERIFIED-TRUE: Section 3.1 reports only bind rate and mean contribution, no component AUC or rank ablation.	Disposition FIX claim; SHELF ablation if runtime is over 30 minutes.	
	The method-identity proof does not cover almost all evaluated timesteps.		FIX contribution and abstract; SHELF truncated-regime analysis.
Class Factual	Verification VERIFIED-TRUE: Section 3.2 limits exact pinning below cap and says evaluated runs spend nearly all time in the truncated regime.	Disposition FIX contribution and abstract; SHELF truncated-regime analysis.	

	"AUC-ROC 0.50962 means constant" is false.		FIX
Class Factual	Verification VERIFIED-TRUE: Table 5 reports 747 distinct scores and nonzero standard deviation.	Disposition FIX: "near-chance ranking on this sample."	
	Interleaving does not manufacture exchangeability.		FIX Sections 2 and 4.3.
Class Factual/statistical	Verification VERIFIED-TRUE: exchangeability is a distributional permutation symmetry; reordering one realization does not establish it, and cited post-2005 literature handles non-exchangeability.	Disposition FIX Sections 2 and 4.3.	
	The repair result is one untuned configuration, not a general textbook repair result.		FIX wording; SHELF broader validation.
Class Factual	Verification VERIFIED-TRUE: Section 7 specifies one stream, one prefix, one prior, and no tuning.	Disposition FIX wording; SHELF broader validation.	
	The prevalence sweep changes more than prevalence and additive lift has unequal headroom.		FIX interpretation/metric display; SHELF matched or weighted rerun.
Class Factual/design	Verification VERIFIED-TRUE: Section 6 says resampled levels do not share identical records; maximum additive lift is 1-p.	Disposition FIX interpretation/metric display; SHELF matched or weighted rerun.	
	LITNET pooled prevalence is an equal-weight identity.		KEEP, but label as arithmetic audit utility.
Class Factual	Verification VERIFIED-TRUE: $(15.7747\% + 3.544\% + 0.176\%)/3 = 6.498233\%$, and attacks sum to 14,621.	Disposition KEEP, but label as arithmetic audit utility.	
	Public CALIBURN results overlap the current result set.		FIX disclosure before submission; notify editor with overlap matrix.
Class Factual	Verification VERIFIED-TRUE: CALIBURN Tables 4-5/Figure 5 report the same rounded pooled LITNET and interleaved CiCIDS method values.	Disposition FIX disclosure before submission; notify editor with overlap matrix.	

The provenance system proves correctness.			No accusation; FIX wording only if any broader "verified" language implies correctness.
Class Factual claim attributed to manuscript	Verification VERIFIED-FALSE as a general inference: it proves traceability/integrity only; the manuscript does not literally claim hashes alone prove scientific correctness.	Disposition No accusation; FIX wording only if any broader "verified" language implies correctness.	
The artifact and eighteen incidents are auditable from this packet.			SHELF pending artifact review; ensure anonymous artifact is present at submission.
Class Factual	Verification UNVERIFIABLE: the artifact, ledger, manifests, and incident file were not supplied.	Disposition SHELF pending artifact review; ensure anonymous artifact is present at submission.	

Response-shelf additions from this fresh round

F1. "The conceptual novelty is already in Catillo and Arp."

Prepared response. We agree that arbitrary ordering, sampling bias, and temporal snooping are known concerns. We will narrow the novelty claim. The contribution is the exact same-full-multiset intervention on a real IDS benchmark, with held-out overlap, relocation, prevalence, and deterministic method ordering measured together, plus the algebraic audit of the released detector. We will no longer describe this as orthogonal to prior benchmark criticism.

Evidence. Catillo et al., pp. 3-4; Arp et al., Sections 2.1-2.2; manuscript Sections 5.1-5.3.

F2. "The paper does not isolate an order effect on detector quality."

Prepared response. Correct. The current manipulation estimates the effect of the complete stream-construction pipeline under a fixed positional split. It changes membership, prevalence, and order within the splits. We will state that estimand explicitly and remove language implying that the same held-out records were scored in both arms. A factorial decomposition of training order, test order, membership, and prevalence is revision-window work.

Evidence. Manuscript Sections 5.1-5.3; 32.5% held-out overlap.

F3. "The method-identity proof does not cover the truncated regime."

Prepared response. Correct. Exact hazard pinning is proved only before truncation, while the implementation's later recurrence must be characterized separately. We will narrow the algebraic claim, distinguish $P(r=0)$ from the scored $P(r \leq 5)$, and remove the assertion that the tail term performs all discrimination unless a component ablation supports it.

Evidence. Manuscript Sections 3.1-3.2.

F4. "The companion papers do not have disjoint result sets."

Prepared response. Correct. The public companion reports rounded versions of the pooled LITNET and interleaved CICIDS results retained here. The new paper's contribution is the corrected interpretation and the natural-versus-constructed contrast, not disjoint underlying measurements. We will replace the sentence with a table identifying reused, re-derived, corrected, and new results and provide the editor with version identifiers and correction status.

Evidence. Manuscript Tables 3-4 and Companion Paper Disclosure; public CALIBURN Tables 4-5 and Figure 5.

F5. "Natural chronology is not universally the right stream."

Prepared response. Agreed. The recommendation is not to privilege chronology by fiat. It is to preserve source/time structure, justify every assembly, and report sensitivity to plausible constructions. For CICIDS2017, timestamp order is the least transformed available reference; for LITNET, no global chronology exists and per-capture reporting is therefore the defensible reference.

Evidence. Manuscript Sections 4.1-4.3, 5.4, and 9.

Sources checked

1. ACM, "Digital Threats: Research and Practice" home and charter. <https://dl.acm.org/journal/dtrap> and <https://dl.acm.org/journal/dtrap/charter> (accessed 26 August 2026).
2. ACM, "DTRAP Author Guidelines." <https://dl.acm.org/journal/dtrap/author-guidelines> (accessed 26 August 2026).

3. Marta Catillo, Antonio Pecchia, and Umberto Villano, "Machine Learning on Public Intrusion Datasets: Academic Hype or Concrete Advances in NIDS?" IEEE/IFIP DSN-S, 2023. <https://doi.org/10.1109/DSN-S58398.2023.00038>. Author PDF: <https://ultraviolet.ding.unisannio.it/villano/papers/DSN2023.pdf>.
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Research stop note

The search stopped after the required answer slots were supported: DTRAP scope, prior-art boundary, exact-paper arithmetic, method-identity theory, exchangeability terminology, public companion overlap, and current ACM guidance. Additional searches returned broader or weaker examples of temporal-order sensitivity but did not alter the verdict. The unresolved gaps require the project's private artifact or new experiments, not more web retrieval.